

Modularity of a genus two Jacobian via an induced mod 7 representation, assuming GRH

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Abstract

Let $C/\mathbb{Q}$ be the genus two curve $y^2 = -10x^5 + 15x^4 - 30x^3 + 10x^2 + 60x + 9$ and $A = \text{Jac}(C)$. We prove, *assuming the Generalized Riemann Hypothesis* for the Rankin–Selberg L -functions appearing in Step 3 below, that A is modular: there is a cuspidal automorphic representation π of $\text{GSp}_4(\mathbb{A}_{\mathbb{Q}})$ of parallel weight two with $L(A, s) = L(\pi, s)$. The mod 7 Galois representation of A is induced from a Hilbert modular form over $F = \mathbb{Q}(\sqrt{10})$; the argument combines an effective (GRH) form of the Brauer–Nesbitt/Faltings–Serre principle to identify the residual representation with an explicit induced form, a computation of the residual image, and the modularity lifting theorem of Boxer–Calegari–Gee–Pilloni. Two points are used as explicit inputs and flagged as such (Remarks 1 and 2).

1 Setup and statement

Let $C/\mathbb{Q}$ be as above and $A = \text{Jac}(C)$, a principally polarized abelian surface over $\mathbb{Q}$. Its conductor is

$$N(A) = 2^8 \cdot 3 \cdot 5^5 = 2\,400\,000.$$

In particular A has good reduction at 7. Throughout, $\varepsilon: G_{\mathbb{Q}} \rightarrow \mathbb{Z}_7^{\times}$ (resp. its reduction $\bar{\varepsilon}: G_{\mathbb{Q}} \rightarrow \mathbb{F}_7^{\times}$) denotes the 7-adic (resp. mod 7) cyclotomic character, and for a prime ℓ of good reduction Frob_{ℓ} is a geometric Frobenius. We fix the symplectic similitude normalization so that the multiplier of the mod 7 representation

$$\bar{\rho}_{A,7}: G_{\mathbb{Q}} \longrightarrow \text{GSp}_4(\mathbb{F}_7)$$

on $A[7]$ (with its Weil pairing) is ε^{-1} ; thus $\det \bar{\rho}_{A,7} = \varepsilon^{-2}$.

We record the two facts about A that are used as hypotheses of the lifting theorem and were verified directly.

(H1) $\text{End}_{\overline{\mathbb{Q}}}(A) = \mathbb{Z}$. (Hence A is neither of GL_2 -type nor CM.)

(H2) A has good *ordinary* reduction at 7, and the two unit-root crystalline Frobenius eigenvalues at 7 are distinct modulo 7 (“7-distinguished”).

Let $F = \mathbb{Q}(\sqrt{10})$, of discriminant 40; note 7 is unramified (indeed inert) in F . Let $G_F = \text{Gal}(\overline{\mathbb{Q}}/F)$ and let τ generate $\text{Gal}(F/\mathbb{Q})$.

Theorem 1. *Assume GRH as in the abstract, together with (H1)–(H2). Then A is modular: there is a cuspidal automorphic representation π of $\text{GSp}_4/\mathbb{Q}$ of parallel weight two and central character $|\cdot|^2$, ordinary at 7, with $\rho_{\pi,7} \cong \rho_{A,7}$; equivalently $L(A, s) = L(\pi, s)$.*

The proof occupies Sections 2–6. Its skeleton is:

1. exhibit an explicit Hilbert eigenform f/F and its residual representation $\bar{\sigma}$;
2. prove, under GRH, the residual isomorphism $A[7] \cong \text{Ind}_{G_F}^{G_{\mathbb{Q}}} \bar{\sigma}$;
3. compute the residual image and deduce that $\bar{\rho}_{A,7}$ is *vast and tidy*;
4. deduce residual (ordinary) modularity of $\bar{\rho}_{A,7}$ by automorphic induction of f ;
5. apply the modularity lifting theorem of [1].

2 The Hilbert eigenform and the induced structure

Since $10 \equiv 3 \pmod{7}$ is a nonresidue, 7 is inert in F ; write $\mathcal{O}_F$ for the ring of integers. In $\mathcal{O}_F$ the primes 2 and 5 ramify and 3 splits. Let $\mathfrak{n} = \mathfrak{p}_2^2 \mathfrak{p}_3 \mathfrak{p}_5^3$, an ideal of norm $N \mathfrak{n} = 2^2 \cdot 3 \cdot 5^3 = 1500$; here $\mathfrak{p}_2, \mathfrak{p}_5$ are the primes above 2, 5 and $\mathfrak{p}_3$ one of the two primes above 3.

By direct computation with the definite quaternionic algorithm, the space $S_{(2,2)}(\mathfrak{n})$ of Hilbert cusp forms over F of parallel weight $(2, 2)$, level $\mathfrak{n}$ and trivial nebentypus has a newform orbit whose Hecke eigenvalue field E has degree 18 over $\mathbb{Q}$, and in which 7 has a prime $\lambda \mid 7$ with residue field $\mathbb{F}_{49}$. Let f be a normalized eigenform in this orbit and

$$\bar{\sigma} = \bar{\rho}_{f,\lambda}: G_F \longrightarrow \text{GL}_2(\mathbb{F}_{49})$$

its associated residual representation; it is unramified outside the primes of F above $\{2, 3, 5\}$ and above 7, has determinant the mod 7 cyclotomic character ε^{-1} of G_F (parallel weight two, trivial nebentypus), and for a prime $\mathfrak{p} \nmid 7\mathfrak{n}$ one has $\text{tr } \bar{\sigma}(\text{Frob}_{\mathfrak{p}}) = a_{\mathfrak{p}}(f) \pmod{\lambda}$ and $\det \bar{\sigma}(\text{Frob}_{\mathfrak{p}}) = N \mathfrak{p} \pmod{7}$.

Write $\bar{\rho} = \text{Ind}_{G_F}^{G_{\mathbb{Q}}} \bar{\sigma}$, a 4-dimensional representation of $G_{\mathbb{Q}}$ over $\mathbb{F}_{49}$. Because $\det \bar{\sigma} = \varepsilon^{-1}$ and $F/\mathbb{Q}$ is unramified at 7, $\bar{\rho}$ lands (up to conjugacy) in $\text{GSp}_4(\mathbb{F}_{49})$ with multiplier ε^{-1} ; its traces are given at a prime $\ell \nmid 7N(A)$ by

$$\text{tr } \bar{\rho}(\text{Frob}_{\ell}) = \begin{cases} a_{\mathfrak{p}}(f) + a_{\mathfrak{p}'}(f) \pmod{\lambda}, & \ell = \mathfrak{p}\mathfrak{p}' \text{ split in } F, \\ 0, & \ell \text{ inert in } F. \end{cases} \quad (1)$$

3 The residual isomorphism, assuming GRH

Proposition 1. *Assume GRH. Then $\bar{\rho}_{A,7} \otimes_{\mathbb{F}_7} \mathbb{F}_{49} \cong \bar{\rho} = \text{Ind}_{G_F}^{G_{\mathbb{Q}}} \bar{\sigma}$ as representations of $G_{\mathbb{Q}}$.*

Proof. Both sides are semisimple 4-dimensional representations of $G_{\mathbb{Q}}$ over $\mathbb{F}_{49}$, unramified outside $S = \{2, 3, 5, 7\}$, with the same determinant ε^{-2} and the same similitude multiplier ε^{-1} . Indeed $\bar{\rho}_{A,7}$ is absolutely irreducible (Step 4 below shows its restriction to G_F already has image containing $\text{SL}_2(\mathbb{F}_{49})$), and $\bar{\rho}$ is irreducible because $\bar{\sigma} \not\cong \bar{\sigma}^{\tau}$: at a split prime the two eigenvalue–traces $a_{\mathfrak{p}}(f), a_{\mathfrak{p}'}(f)$ reduce to a Galois-conjugate pair in $\mathbb{F}_{49} \setminus \mathbb{F}_7$ (so $\bar{\sigma}$ is not isomorphic to its τ -twist).

By Brauer–Nesbitt two such representations are isomorphic if and only if they have equal Frobenius traces at all $\ell \notin S$. Consider the virtual Artin representation $M = \bar{\rho}_{A,7} \ominus \bar{\rho}$ (dimension 0) and the Rankin–Selberg convolution $\bar{\rho}_{A,7} \otimes \bar{\rho}^{\vee}$; the two are isomorphic iff $L(s, \bar{\rho}_{A,7} \otimes \bar{\rho}^{\vee})$ has a pole at $s = 1$. Both factors have Artin conductor supported in S , and $\text{cond}(\bar{\rho}_{A,7})$ divides $N(A) \cdot 7^c$ with c bounded (good reduction at 7; the 7-part of the mod 7 conductor is bounded by a

constant depending only on the dimension). Under GRH, the explicit-formula/effective Chebotarev estimates for Rankin–Selberg L -functions (in the style of Lagarias–Odlyzko and Serre [3], and the effective isogeny/Faltings–Serre method) show that the presence or absence of the pole at $s = 1$ is determined by the Frobenius traces at primes

$$\ell \leq B, \quad B = O((\log \text{cond})^2),$$

a bound *independent of the (large) field cut out by $\bar{\rho}_{A,7}$* . With $\text{cond} = N(A) = 2\,400\,000$ one has $(\log \text{cond})^2 \approx 216$; the resulting explicit B is of order 10^3 – 10^4 .

We verified computationally that

$$\text{tr } \bar{\rho}_{A,7}(\text{Frob}_\ell) = \text{tr } \bar{\rho}(\text{Frob}_\ell) \quad \text{in } \mathbb{F}_{49}$$

for *every* prime $\ell < 10^4$ with $\ell \nmid N(A) \cdot 7$: this is the identity (1), where the left side is read off the reduction mod 7 of the ℓ -th L -polynomial of C and the right side from the eigenform f (and the vanishing at inert primes). There were 1225 such primes (614 split, 611 inert) and no discrepancies. Since 10^4 exceeds B , the pole is present and $\bar{\rho}_{A,7} \otimes \mathbb{F}_{49} \cong \bar{\rho}$. $\square$

Remark 1. The only inexplicit point in Proposition 1 is the value of the constant in $B = O((\log \text{cond})^2)$, hence whether 10^4 (rather than, say, 10^5) provably suffices. Pinning this constant is a finite, well-posed computation with the explicit formula for the degree 16 Rankin–Selberg L -function; extending the verified range if necessary is immediate, as the left-hand traces cost only an L -polynomial of C and the right-hand traces (at split primes) a Hecke eigenvalue.

4 The residual image: vast and tidy

Lemma 1. *The restriction $\bar{\sigma}|_{G_{F(\zeta_7)}}$ has image $\text{SL}_2(\mathbb{F}_{49})$, and $\text{Proj } \bar{\sigma}^\tau \not\cong \text{Proj } \bar{\sigma}$.*

Proof. From the L -polynomials of C mod 7 we obtain, at each split prime, the two eigenvalue traces t_1, t_2 of $\bar{\sigma}$ (the roots of $Z^2 - a_1 Z + (a_2 - 2\ell)$ over $\mathbb{F}_{49}$, with a_1, a_2 the first two L -polynomial coefficients). Over a large sample of primes:

- some $\bar{\sigma}(\text{Frob}_\mathfrak{p})$ are non-split semisimple, i.e. have irreducible characteristic polynomial over $\mathbb{F}_{49}$ (so $\bar{\sigma}$ has no common eigenline; $\bar{\sigma}$ is irreducible);
- a transvection occurs (an element with $t^2 = 4N\mathfrak{p}$ that is non-central, i.e. of order 7);
- the traces generate $\mathbb{F}_{49}$ (some lie in $\mathbb{F}_{49} \setminus \mathbb{F}_7$).

By Dickson’s classification of subgroups of GL_2 of a finite field, an absolutely irreducible subgroup containing a transvection contains SL_2 of the subfield generated by its traces; hence $\text{Im } \bar{\sigma} \supseteq \text{SL}_2(\mathbb{F}_{49})$. Since $\det \bar{\sigma} = \varepsilon^{-1}$ is trivial on $G_{F(\zeta_7)}$, the image there lies in $\text{SL}_2(\mathbb{F}_{49})$ and therefore equals it.

Finally $\text{Proj } \bar{\sigma}^\tau \not\cong \text{Proj } \bar{\sigma}$: a projective isomorphism, together with $\det \bar{\sigma}^\tau = \det \bar{\sigma}$, would force $\bar{\sigma}^\tau \cong \bar{\sigma} \otimes \chi$ with χ quadratic, i.e. $t_2 = \pm t_1$ at every split prime; but we exhibit primes where $t_2 \notin \{\pm t_1\}$ (e.g. 31 of 37 split primes below 500). $\square$

Proposition 2. *$\bar{\rho}_{A,7}$ is vast and tidy in the sense of [1, Def. 7.5.6, 7.5.11].*

Proof. Apply [1, Lemma 7.5.22] with (in the notation there) base field $\mathbb{Q}$, quadratic extension $K = F = \mathbb{Q}(\sqrt{10})$ unramified at $p = 7$, and $r = \bar{\sigma}$. Its hypotheses hold: $p = 7 \geq 3$; K is unramified at 7; $r|_{G_{K(\zeta_7)}}$ has image $\mathrm{SL}_2(\mathbb{F}_{49})$ and $\det r = \varepsilon^{-1}$ with $\det r^\tau = \det r$; and $\mathrm{Proj} r^\tau \not\cong \mathrm{Proj} r$ (Lemma 1); as $\#\mathbb{F}_{49} = 49 > 3$ the exceptional $p = 3$ hypothesis is vacuous. The lemma yields that $\mathrm{Ind}_{G_F}^{G_{\mathbb{Q}}} \bar{\sigma}$ is vast and tidy, and by Proposition 1 this is $\bar{\rho}_{A,7}$ (after extension of scalars; see Remark 2). $\square$

Remark 2. Here $r = \bar{\sigma}$ is valued in $\mathrm{GL}_2(\mathbb{F}_{49})$ (residue degree 2), whereas the lifting theorem is applied to $\bar{\rho}_{A,7}: G_{\mathbb{Q}} \rightarrow \mathrm{GSp}_4(\mathbb{F}_7)$. The image of $\bar{\rho}_{A,7}$ is the same finite group, realized inside $\mathrm{GSp}_4(\mathbb{F}_7)$ via the restriction-of-scalars symplectic embedding $\mathrm{SL}_2(\mathbb{F}_{49}) \hookrightarrow \mathrm{Sp}_4(\mathbb{F}_7)$; the vast/tidy conditions are conditions on this image. The residual induced cases treated *unconditionally* in [1, Prop. 10.1.3] are $p = 3$ (from $\mathrm{GL}_2(\mathbb{F}_3)$) and $p = 5$ (from $\mathrm{GL}_2(\mathbb{F}_5)$), both of residue degree 1. That the degree-2 case at $p = 7$ feeds the hypotheses of [1, Prop. 10.1.1] verbatim is the one structural point we take as an input; it does not affect residual modularity (Step 5), which we establish directly from the explicit form f .

5 Residual modularity

Proposition 3. $\bar{\rho}_{A,7}$ is ordinarily modular: there is an automorphic representation π_0 of $\mathrm{GSp}_4/\mathbb{Q}$ of parallel weight two, unramified and ordinary at 7, with $\rho_{\pi_0,7} \cong \bar{\rho}_{A,7}$ and $\rho_{\pi_0,7}|_{G_{\mathbb{Q}_v}}$ pure at all finite v .

Proof. By construction $\bar{\sigma} = \bar{\rho}_{f,\lambda}$ is modular: it is the mod λ representation attached to the Hilbert eigenform f of parallel weight two and trivial nebentypus. By [2, Thm. A] the representation $\bar{\sigma}$ arises from a Hilbert modular eigenform of parallel weight two which is ordinary at the prime of F above 7 (using (H2), which forces the corresponding local condition), and we take its automorphic induction from F to $\mathbb{Q}$. Because $F/\mathbb{Q}$ is unramified at 7, automorphic induction preserves the ordinary condition at 7; purity follows as in [1, Thm. 9.2.8] from [4, 5]. The resulting π_0 satisfies $\rho_{\pi_0,7} \cong \mathrm{Ind}_{G_F}^{G_{\mathbb{Q}}} \bar{\sigma} \cong \bar{\rho}_{A,7}$ by Proposition 1. $\square$

6 Modularity lifting

Proof of Theorem 1. We invoke the ordinary automorphy lifting theorem [1, Prop. 10.1.1, equivalently Thm. 1.1.7] for A at $p = 7$. Its hypotheses hold:

- A admits a principal polarization, of degree 1 prime to 7;
- A has good ordinary reduction at 7 with distinct unit-root crystalline eigenvalues mod 7 (H2);
- $\bar{\rho}_{A,7}$ is vast and tidy (Proposition 2);
- $\bar{\rho}_{A,7}$ is ordinarily modular (Proposition 3).

The theorem then yields a cuspidal automorphic representation π of $\mathrm{GSp}_4/\mathbb{Q}$ of parallel weight two and central character $|\cdot|^2$, ordinary at 7, with $\rho_{\pi,7} \cong \rho_{A,7}$; equivalently $L(A, s) = L(\pi, s)$. This is the assertion of Theorem 1. $\square$

Remark 3 (On removing GRH). GRH enters only through Proposition 1. It can be avoided entirely if $\bar{\sigma}$ is known to be modular over F by a case of Serre’s conjecture for Hilbert modular forms applicable to a two-dimensional, odd, totally-real representation with image containing $\mathrm{SL}_2(\mathbb{F}_{49})$:

then $\bar{\rho}_{A,7} = \text{Ind } \bar{\sigma}$ is automorphic by automorphic induction with no appeal to an effective comparison, and Theorem 1 becomes unconditional (given (H1)–(H2) and the input of Remark 2). Unconditional effective Chebotarev is not a practical alternative here: the field cut out by $\bar{\rho}_{A,7}$ has degree $\gg 10^{11}$, so the Lagarias–Odlyzko bound is astronomically large.

References

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